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Influence of Water Occurrence on CO₂ Mass - Transfer, Diffusion, Oil Recovery, and Sequestration Mechanisms in Shale Reservoirs Under Coupled Thermal–Mass Effects

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Abstract

The presence of residual fracturing fluid and formation water in shale reservoirs significantly influences CO₂ mass transfer, diffusion, and its mechanisms for enhanced oil recovery and storage. However, the impact of varying water saturation levels and coupled thermal-hydraulic effects on CO₂ EOR and storage mechanisms remains unclear. This study investigates CO₂ mass transfer and diffusion behavior through pressure-driven core flooding and high-temperature huff-and-puff experiments, complemented by NMR analysis, under varying temperatures and water saturation conditions in shale reservoirs. Results show that elevated temperatures shorten the initiation period of oil mobilization, accelerate CO₂ diffusion, and enable the system to reach equilibrium more rapidly. Meanwhile, retained fracturing fluids and formation water create an aqueous barrier that slows down pressure drop rates as water saturation increases. At lower temperatures (<110 °C), oil recovery improvements are limited; however, at 140 °C, oil recovery increases linearly, indicating that CO₂ exhibits both efficient displacement and favorable sequestration characteristics under these conditions. Although early-stage oil recovery is lower under aqueous conditions, subsequent cycles yield more sustained improvements. The presence of water amplifies the incremental benefits of CO₂ huff-and-puff, ultimately enhancing oil recovery by 9.53% compared to dry conditions. This study provides new theoretical insights for optimizing CO₂-based EOR and sequestration strategies in shale reservoirs.

Keywords: Shale Reservoir, Thermal-Mass Effect, CO₂ Huff-and-Puff, Mass Transfer and Diffusion, EOR and Sequestration Mechanisms

Introduction

With growing energy demand and increasing environmental concerns, the development of unconventional oil and gas resources—particularly shale oil and gas—has become a major research focus (Vengosh et al. 2014, Wang et al. 2014, Yan et al. 2017). CO₂-based enhanced oil recovery (EOR) and sequestration technologies in shale reservoirs have attracted significant attention due to their dual benefits of improving

recovery and reducing greenhouse gas emissions (Ning and Tura 2023, Nsiama et al. 2024). However, enhancing the efficiency of CO₂-driven oil recovery and storage in shale formations remains a critical challenge (Tian et al. 2024, Xie et al. 2023).

Inspired by in-situ upgrading of oil shale, some researchers have proposed integrating thermal stimulation with CO₂ injection to improve both reservoir quality and fluid properties, thereby enhancing oil recovery (Alafnan et al. 2020, Li et al. 2014, Zhao et al. 2024). Studies indicate that increasing pyrolysis temperature and organic content can significantly enhance shale permeability and fluid flow behavior when CO₂ is injected (Lu et al. 2021, Shi et al. 2024, Wan et al. 2024). Permeability tends to increase with higher pyrolysis temperatures, and thermal conductivity varies significantly with respect to bedding orientation (Goodman et al. 2021). Additionally, organic matter decomposition generates macro-pores, which may either enhance fluid flow or remain isolated and thus underutilized, affecting overall recovery efficiency. Thermal stimulation also enhances the mass transfer and diffusion capacity of CO₂ within shale formations (Nikkhou et al. 2015, Yang and Gu 2006), but this is a complex process influenced by factors such as pore structure, organic content, and the presence of aqueous phases (Zheng et al. 2016). Currently, molecular dynamics simulations and pore network models are widely employed to investigate CO₂ diffusion in both organic and inorganic nanopores of shale (Li, Lim, et al. 2019, Zhao et al. 2016). Findings suggest that kerogen surface roughness and type have a significant impact on CO₂ diffusion coefficients (Sui et al. 2020). Moreover, the presence of water in shale pores has been shown to reduce the diffusion coefficients of both hydrocarbons and CO₂, underscoring the importance of accounting for water saturation in diffusion studies (Nunn et al. 2018). However, further work is needed to scale these microscopic diffusion insights to reservoir-level applications. At the core scale, numerous studies have identified gas diffusion, interfacial tension reduction, and light hydrocarbon extraction as the primary mechanisms behind CO₂-driven oil recovery (Pu et al. 2016, Song and Yang 2017, Zuloaga et al. 2017). In recent years, researchers have conducted in-depth studies on the performance of CO₂ injection across various shale reservoir types. These studies reveal that CO₂ huff-and-puff techniques can significantly enhance shale oil and gas recovery while also enabling effective CO₂ adsorption and sequestration (Guo et al. 2025, Xia et al. 2024). Matrix permeability plays a critical role in improving CO₂ extraction efficiency, especially in formations with larger pores. Overall, these findings suggest that optimizing CO₂ huff-and-puff operations can maximize hydrocarbon recovery and CO₂ storage efficiency across diverse reservoir conditions.

In summary, although substantial progress has been made in thermal stimulation, CO₂ mass transfer, and EOR-sequestration mechanisms in shale, many uncertainties remain regarding how temperature and water saturation influence CO₂ diffusion characteristics and recovery/sequestration performance. This study focuses on core-scale experiments to examine how temperature and aqueous phase conditions affect CO₂ mass transfer, diffusion, and EOR-sequestration mechanisms in shale reservoirs. The aim is to provide new theoretical guidance for developing more effective CO₂-based EOR and storage strategies.

Experiment and Materials

Experimental Samples

The shale samples used in this study were obtained from actual cores taken from Block L of the Liaohe Oilfield. The petrophysical properties of the samples and the corresponding experimental details are summarized in Table 1. For core cleaning, a mixture of petroleum ether and ethanol was used as the organic solvent. The petroleum ether was supplied by Tedia High Purity Solvents (Anhui, China), and the ethanol was obtained from Sinopharm Chemical Reagent Co., Ltd. Simulated oil was prepared by blending degassed formation crude oil with kerosene, the latter supplied by Macklin Biochemical Co., Ltd. (Shanghai, China). To enable nuclear magnetic resonance (NMR) measurements required in the thermally assisted CO₂ huff-and-puff experiments, the water phase was formulated using heavy water (from Wuhan Isotope Technology

Co., Ltd.) diluted at a ratio of 1% with a polymer solution containing polyquaternium. For the convection-diffusion experiments, the aqueous phase was similarly prepared using deionized water and polyquaternium at the same dilution ratio. The CO₂ gas used throughout all experiments was supplied by Qingdao Xinkeyuan Gas Co., Ltd., with a purity of 99.99%.

Table 1—Core Physical Properties and Experimental Summary.

Core No.	Length (cm)	Diameter (cm)	Dry weight (g)	Porosity (%)	Permeability (mD)	Experiment
1	4.94	2.48	61.81	6.93	40.575	CO ₂ convection-diffusion experiment
2	5.95	2.50	68.26	7.07	113.154	CO ₂ huff and puff experiment (temperature)
3	4.74	2.53	58.35	6.87	43.665	CO ₂ huff and puff experiment (water occurrence)

Experimental Equipment

All experimental apparatuses used in this study were independently designed and assembled. A schematic diagram of the CO₂ convection-diffusion experimental setup is shown in Figure 1, while Figure 2 illustrates the configuration for the thermally assisted CO₂ huff-and-puff experiments. The key instruments employed include a MacroMR12-110H-I nuclear magnetic resonance (NMR) analyzer, a constant-temperature oven, a core holder, a high-temperature and high-pressure intermediate vessel, a gas booster pump, a hand pump, and a Vindum displacement pump. For core sample preparation, the main equipment included a core cleaning device, a drying oven, and a shale porosity and permeability measurement system.

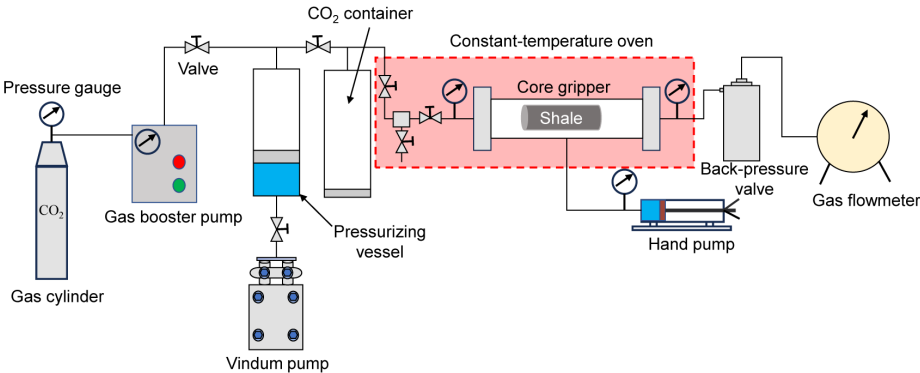


Figure 1—Schematic diagram of the CO₂ convection-diffusion experimental apparatus.

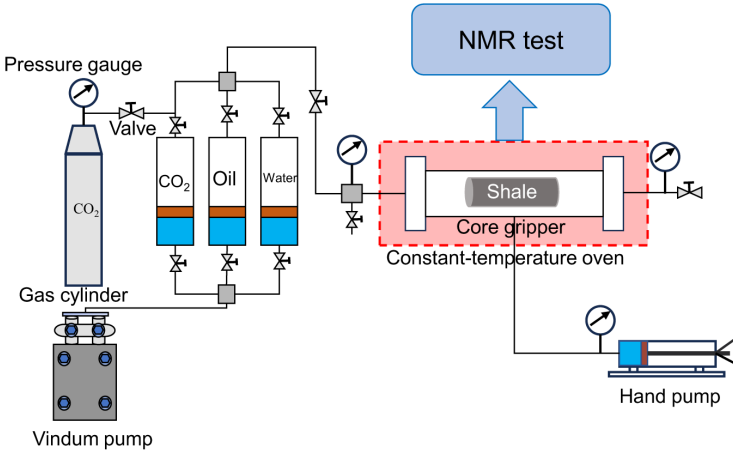


Figure 2—Schematic diagram of the thermally assisted CO₂ huff-and-puff experimental setup.

Experimental Procedure

CO₂ Convection-Diffusion Experimental Procedure in Porous Media. This experiment simulates the convection-diffusion of CO₂ during the soaking stage of huff-and-puff processes, where CO₂ is driven by pressure into the shale matrix. The main procedures are as follows:

1. Both ends of the shale core were trimmed flat using wire-cutting to ensure smooth surfaces and prevent stress concentration or core failure during the experiment.
2. The core was soaked in a 1:1 volume ratio mixture of ethanol and petroleum ether using a core cleaning apparatus. The cleaning was conducted at 100 °C for 7 days.
3. The core was dried in a constant-temperature oven at 100 °C for 24 hours to eliminate residual solvents and moisture. Basic petrophysical properties were then measured, as listed in [Table 1](#).
4. The core was saturated with simulated oil using pressurized saturation at 20 MPa for 7 days.
5. The saturated core was placed in a core holder with a backpressure valve set to 20 MPa. A displacement pump injected dodecane at 20 MPa for 12 hours to ensure internal pressure equilibrium within the core.
6. CO₂ was pressurized to 30 MPa and injected into a high-temperature, high-pressure intermediate vessel (maintained at room temperature). The constant-temperature oven was then set to the target experimental temperatures (80 °C, 100 °C, 120 °C, and 140 °C). Once the system stabilized, the inlet valve was opened to initiate CO₂ injection. Real-time pressure variations at the inlet and outlet of the core were recorded.
7. After pressure equilibrium was reached (or a preset time elapsed), CO₂ injection was stopped by closing the inlet valve. The system pressure was gradually reduced in 2 MPa increments, holding each step for 5 minutes to avoid stress-induced damage. Meanwhile, a vent valve was used to safely release the system pressure to atmospheric levels.

Thermally Assisted CO₂ Huff-and-Puff Experimental Procedure. This experiment simulates the field-relevant CO₂ huff-and-puff cycle, consisting of injection, soaking, and backflow stages. The injection pressure was set to 15 MPa, and the test parameters considered both water saturation and temperature conditions. The procedures are as follows:

1. The cleaning, petrophysical measurements, and oil saturation steps were consistent with those described in the convection-diffusion procedure.
2. NMR measurements were performed on the oil-saturated core to obtain the initial T₂ relaxation curve.
3. The saturated core was placed into the core holder, and the outlet valve was closed. CO₂ was injected at a constant pressure until inlet and outlet pressures both reached 15 MPa.
4. The inlet valve was then closed to initiate the soaking phase, which lasted 3 hours. Simultaneously, the oven was turned on to raise the system temperature to the experimental setpoint. During this phase, pressure variations at both ends were continuously monitored.
5. After soaking, the core was depressurized via the vent valve, and once cooled to room temperature, a post-soaking T₂ spectrum was recorded. This completed the first huff-and-puff cycle.
6. Steps (3)–(5) were repeated for a total of four huff-and-puff cycles.
7. Finally, the core was subjected to T₂ relaxation NMR measurements to determine the distribution of residual oil after thermally assisted CO₂ huff-and-puff.

Results and Discussion

Effect of Temperature and Aqueous Phase Presence on CO₂ Convection-Diffusion Behavior

To investigate the effect of temperature on CO₂ convection-diffusion behavior in porous media, a series of experiments were conducted under a simulated pressure differential of approximately 11 MPa, with the upstream pressure maintained at 31 MPa and the downstream at 20 MPa. The diffusion temperatures were set at 80 °C, 100 °C, 120 °C, and 140 °C. As shown in Figure 3, the pressure decay curves at both ends of the core illustrate distinct variations under different thermal conditions. Based on the time-dependent pressure drop, the CO₂ mass transfer and diffusion process can be conceptually divided into four stages. In the initial interfacial dissolution stage, pressure changes slowly, indicating that the system is in its activation phase. At this point, due to interfacial tension, capillary forces, and delayed solubility equilibrium, CO₂ dissolves primarily at the inlet interface, forming a localized concentration gradient near the core entrance without significant inward diffusion. As the process progresses into the convection-diffusion phase, the pressure drops rapidly, suggesting efficient CO₂ penetration into the core accompanied by a large concentration gradient and enhanced mass transfer driven by pressure-induced convection. Following this, during the gradient attenuation phase, the pressure decline rate decreases linearly as the diffusion front approaches the outlet. The reduced concentration gradient, increased CO₂ saturation, and extended diffusion path collectively lead to higher mass transfer resistance and decreased diffusion rates. Finally, in the quasi-steady near-equilibrium stage, pressure changes become minimal, indicating that CO₂ distribution within the core approaches equilibrium, with a weak residual concentration gradient and nearly constant flux dominated by molecular diffusion. At lower temperatures such as 80 °C, CO₂ exhibits higher solubility in the oil phase but a slower dissolution rate, resulting in a prolonged initiation phase. As temperature increases, CO₂ solubility slightly decreases; however, molecular thermal motion is enhanced, accelerating the dissolution rate and shortening the initial phase duration. During the convection-diffusion phase, higher temperatures further promote faster diffusion and improved mass transfer efficiency, leading to more pronounced pressure declines. Consequently, systems at higher temperatures reach steady-state conditions earlier, entering the subsequent gradient decay and near-equilibrium stages more quickly and ultimately shortening the time required to achieve overall mass transfer equilibrium.

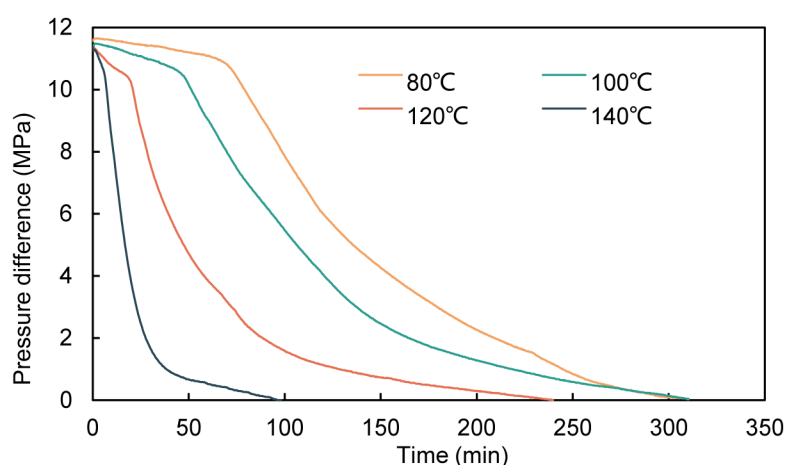


Figure 3—Pressure decay curves across core ends under different diffusion temperatures.

In Figure 4, the pressure decay curves at both ends of the core under varying water saturation conditions are presented. The low water saturation condition refers to cases where the tubing at the core inlet is entirely filled with water, simulating the influence of residual fracturing fluid on subsequent CO₂ mass transfer and diffusion. The medium water saturation condition represents scenarios where the inlet tubing is also filled

with water and the core's internal water saturation is approximately 40%, simulating moderate imbibition of fracturing fluid and the presence of formation water. The high water saturation condition reflects a more complete imbibition process, where the inlet tubing remains fully water-filled and the internal water saturation reaches approximately 80%, representing extensive water presence within the reservoir. Under dry conditions, the core exhibits a steep initial pressure decay, indicating rapid CO₂ diffusion. In contrast, when water is present, the pressure drop rate is significantly reduced. This deceleration is attributed to two key mechanisms: first, the aqueous phase impedes CO₂ diffusion into the oil phase; second, pore water forms thin liquid films that block pore throats and increase the tortuosity of diffusion pathways, thereby enhancing the contribution of Knudsen diffusion (Mo et al. 2022). As water saturation increases, the water phase becomes progressively continuous, shifting the dominant mass transfer mechanism to interfacial gas–liquid dissolution and desorption. This transition introduces additional mass transfer resistance. Overall, the presence of residual fracturing fluid inhibits CO₂ diffusion through a dual mechanism involving both a frontal water barrier and the internal distribution of pore-scale water within the core.

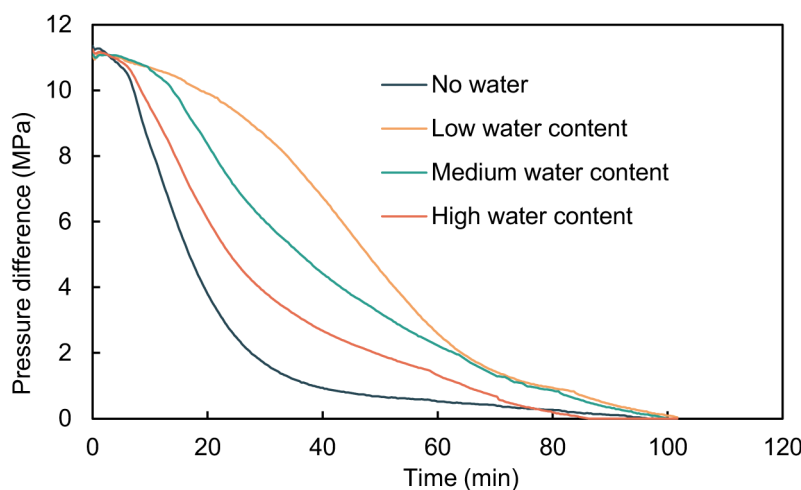


Figure 4—Pressure decay curves across core ends under different water saturation conditions.

Assuming that CO₂ in the container behaves as an ideal gas, the outlet end of the core remains under constant pressure, and the pore volume of the core is much smaller than that of the container, the diffusion process within the core is considered significantly faster than the pressure variation in the container. Thus, the concentration distribution inside the core is assumed to reach a steady state instantaneously and follows a linear profile. Diffusion is assumed to occur only along the axial direction of the core, with a uniform and constant cross-sectional area. Apart from diffusion through the core, no gas leakage or external input occurs within the system. Based on these assumptions, and in combination with Fick's law, the concentration decay process is simplified as a pressure decay process, expressed as follows (Chen et al. 2025):

$$P(t) = P_0 e^{-\frac{D_e A t}{V_g L}} \quad (1)$$

Where $P(t)$ is the time-dependent pressure function, MPa; P_0 is the initial pressure, MPa; D_e is the effective diffusion coefficient of CO₂, cm²/s; A is the cross-sectional area of the core, cm²; t is time, s; V_g is the volume of the CO₂ container, mL; L is the length of the core, cm.

Based on the experimentally recorded pressure decay data over time, the natural logarithm of pressure $\ln P(t)$ was plotted against time t , and a linear fit was applied. The slope of the resulting curve was then used to calculate the effective diffusion coefficient for each stage using the following equation:

$$D_e = -\frac{V_g L m}{A} \quad (2)$$

Here, m is the slope of the linear fit of $\ln P(t)$ versus time t .

The calculated effective diffusion coefficients D_c of CO_2 and the corresponding correlation coefficients R_2 under different temperature and water saturation conditions are presented in Table 2.

Table 2—The stage-wise effective diffusion coefficients of CO_2 and the corresponding R_2 values under varying temperature and water saturation conditions.

Experimental condition		Effective diffusion coefficient ($10^{-4} \text{ cm}^2/\text{s}$)				R_2			
Temperature	Water content	Dissolution stage	Convection-diffusion phase	Gradient attenuation phase	Near-equilibrium stage	Dissolution stage	Convection-diffusion phase	Gradient attenuation phase	Near-equilibrium stage
80°C	No	5.8728	59.24	30.429	16.404	0.9603	0.9994	0.9946	0.958
100°C	No	10.844	59.347	40.147	10.27	0.981	0.9976	0.9839	0.9802
120°C	No	28.808	110.12	42.792	7.8696	0.9887	0.9805	0.9791	0.9798
140°C	No	74.952	296.43	107.24	13.227	0.9688	0.9956	0.9774	0.9763
140°C	Low	37.109	131.29	79.037	35.619	0.9649	0.993	0.987	0.9934
140°C	Medium	26.718	147.44	80.135	36.781	0.8274	0.9957	0.9948	0.9893
140°C	High	50.212	178.05	53.728	30.718	0.9098	0.9863	0.9957	0.9842

Analysis of Temperature Effects on CO_2 -Based Recovery and Sequestration Performance

In this section, a single shale core was subjected to CO_2 huff-and-puff at different temperatures to evaluate the impact of thermal conditions on oil recovery and CO_2 sequestration. A total of four huff-and-puff cycles were conducted under a constant injection pressure of 15 MPa and a soaking time of 3 hours. The T_2 spectra of the core at the original state and after each huff-and-puff cycle are shown in Figure 5. The T_2 spectrum reflects the relaxation time distribution of fluids within the rock pores, with different relaxation times corresponding to fluids in pores of varying sizes. In the initial state, the total T_2 signal represents the quantity of movable fluids originally present in the core. After each huff-and-puff cycle, the total T_2 signal reflects the remaining movable fluid. Therefore, the reduction in signal intensity corresponds to the volume of fluids produced. The degree of recovery after each CO_2 huff-and-puff cycle can be calculated using the following equation:

$$R_i = \frac{S_i - S_0}{S_0} \times 100\% \quad (3)$$

Where R_i is the oil recovery of the i -th CO_2 huff-and-puff cycle, %; S_i is the total T_2 signal after the i -th cycle; S_0 is the total T_2 signal in the original state.

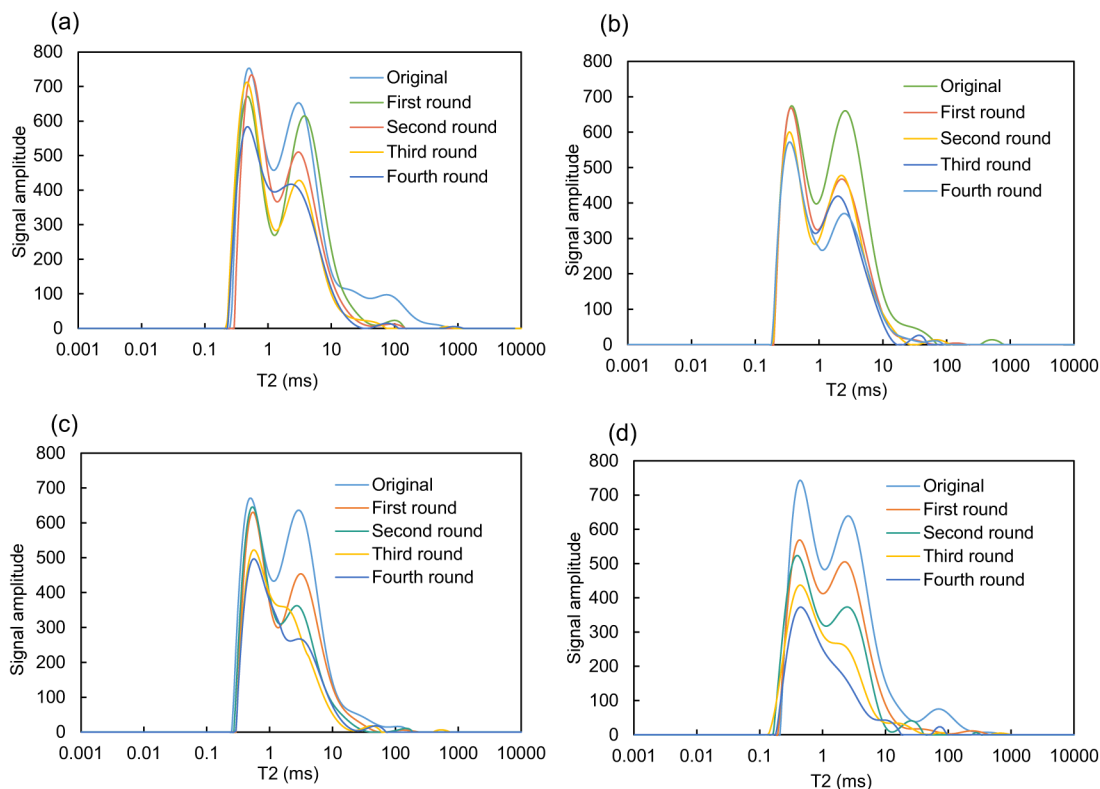


Figure 5— T_2 spectra of the core at the original state and after each CO_2 huff-and-puff cycle under different temperatures: (a) 50 °C, (b) 80 °C, (c) 110 °C, (d) 140 °C.

The oil recovery performance under different temperatures and huff-and-puff cycles was quantitatively assessed, as shown in Figure 6. With increasing huff-and-puff cycles, oil recovery showed a rising trend across all temperature conditions. However, at temperatures below 110 °C, the improvement in recovery beyond the third cycle was marginal. This is attributed to the relatively low diffusion coefficient of CO_2 under low-temperature conditions and its limited interaction with crude oil. Additionally, as the number of cycles increased, the easily mobilized light components were preferentially produced, leaving behind a higher proportion of heavier fractions in the remaining oil. More importantly, the core was subject to boundary effects, which restricted the extent of CO_2 sweep in later cycles, especially in peripheral regions, thus limiting further enhancement in recovery. Higher temperatures resulted in more pronounced recovery improvements, particularly during the later stages of huff-and-puff. At elevated temperatures, the solubility and diffusion coefficient of CO_2 in crude oil increased significantly, leading to reduced oil viscosity and improved flowability. Temperature rise also facilitated the vaporization and extraction of light hydrocarbons, enhancing miscibility between CO_2 and crude oil. Furthermore, high temperatures weakened the resistance of rock surface wettability against oil flow. These synergistic effects enabled CO_2 to displace more oil even in later cycles, thereby extending the effective operational period of the huff-and-puff process under thermal assistance. Among the tested conditions, the best performance was observed at 140 °C, where oil recovery increased almost linearly with each huff-and-puff cycle. In addition, the high density of supercritical CO_2 at this temperature enhanced its capillary trapping potential, further contributing to improved structural sequestration.

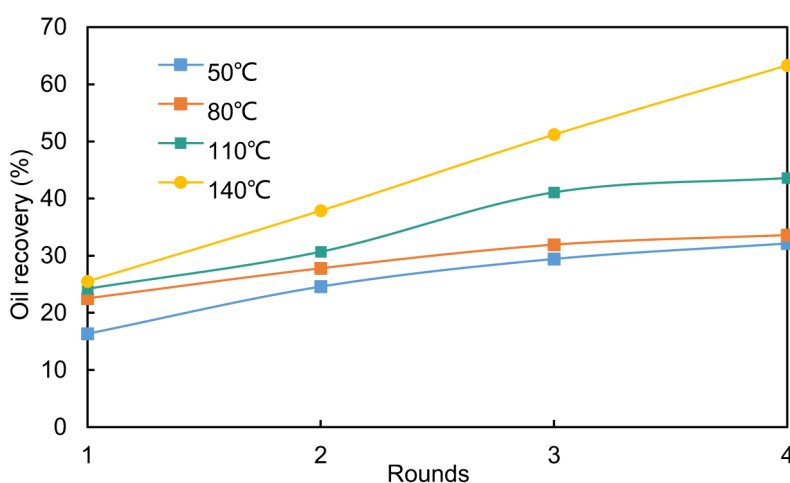


Figure 6—Oil recovery under different temperatures and huff-and-puff cycles, showing the cumulative effect of thermal stimulation.

The comparative effects of temperature elevation and multiple huff-and-puff cycles on CO₂ recovery enhancement are illustrated in Figure 7. During the first huff-and-puff cycle, the most significant improvement in oil recovery was observed within the 50–80 °C temperature range. This is primarily due to the high sensitivity of oil viscosity to temperature changes in this interval; even a slight increase in temperature can lead to a substantial reduction in viscosity. At the same time, the initial solubility of CO₂ in crude oil begins to take effect. However, as the number of cycles increases, the dominant temperature range for recovery enhancement shifts noticeably. In the 110–140 °C range, the recovery increment exhibits an exponential growth trend in later cycles. In contrast, the intermediate temperature range of 80–110 °C shows a gradual, progressive improvement. From the second cycle onward, this range contributes steadily to recovery and maintains a relatively stable enhancement in the later stages. This may indicate that it functions as a transitional zone—possessing moderate thermal activation potential but not fully engaging the deeper mechanisms associated with high-temperature effects. As such, the advantages of high-temperature conditions become more pronounced only after multiple cycles. These findings suggest that in field operations, temperature and huff-and-puff strategies should be dynamically adjusted based on the dominant recovery mechanisms at different stages of the process.

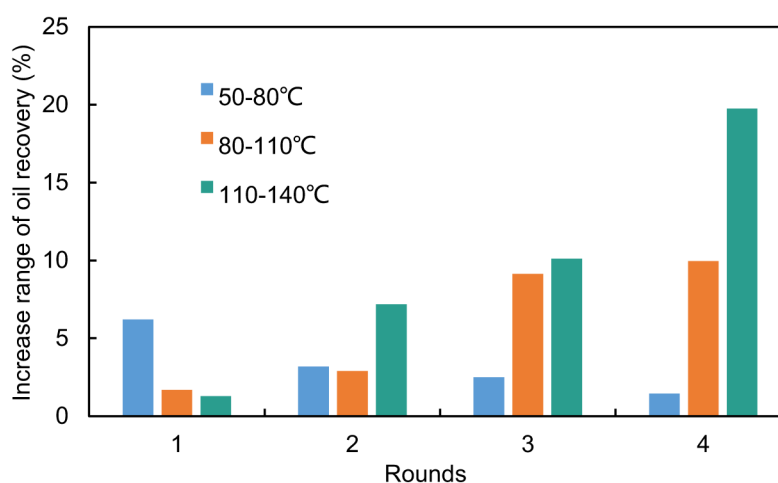


Figure 7—Incremental oil recovery at different temperatures and huff-and-puff cycles, highlighting the shift in dominant temperature effects across cycles.

Based on the T_2 -based classification of shale pore types proposed by Li et al. (Li, Wang, et al. 2019), and in conjunction with the T_2 spectra obtained in this study, the pore size categories were defined as follows: $T_2 < 1$ ms for micropores, $1 \text{ ms} \leq T_2 < 10$ ms for small pores, $10 \text{ ms} \leq T_2 < 100$ ms for medium pores, and $T_2 \geq 100$ ms for macro-pores. The oil recovery from different pore types after four cycles of CO_2 huff-and-puff at varying temperatures is shown in Figure 8. It was observed that for macro-pores, CO_2 injection at all temperatures resulted in nearly complete recovery of movable oil, indicating excellent displacement efficiency. This can be attributed to the high permeability and low capillary resistance in macro-pores, which allow CO_2 to diffuse rapidly and form continuous flow pathways for effective oil displacement. Temperature had a more pronounced effect on oil recovery in small pores. Higher temperatures significantly reduced oil viscosity and enhanced CO_2 solubility and diffusivity, which helped to overcome the higher capillary forces in small pores. Additionally, CO_2 tended to remain longer in small pores under high-temperature conditions, facilitating greater proportions of dissolved-phase sequestration. In micropores, noticeable recovery was only achieved when the temperature exceeded 80°C . This is due to the extremely high capillary pressures and confined space effects in micropores, which require higher temperatures for CO_2 to reach a supercritical state with enhanced solvation and penetration capacity. Moreover, the reduced mobility of supercritical CO_2 in micropores favors longer-term structural sequestration. It is worth noting that at temperatures around 80°C , oil originally present in micropores and small pores can migrate into medium pores due to thermal expansion and displacement by CO_2 . However, since CO_2 has not yet achieved optimal displacement efficiency in medium pores at this stage, a temporary decline in oil recovery from medium pores may occur. As temperature continues to increase beyond 80°C , the recovery from medium pores also improves. Concurrently, the CO_2 storage mechanism in medium pores transitions from primarily free-phase to a combination of dissolved and residual phases, significantly enhancing sequestration security.

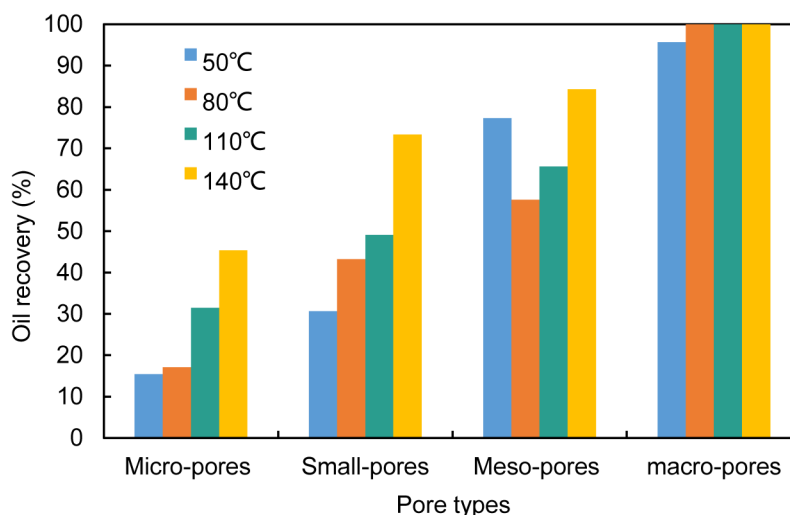


Figure 8—Oil recovery from different pore types after four CO_2 huff-and-puff cycles under varying temperatures.

Effect of water occurrence on CO_2 -Based Recovery and Sequestration Performance

This section investigates the influence of fracturing fluid retention on subsequent CO_2 huff-and-puff recovery and sequestration performance. Specifically, an aqueous-phase huff-and-puff process was first conducted using water containing a clay stabilizer, followed by multiple cycles of CO_2 huff-and-puff at 140°C . The oil recovery under aqueous-phase conditions was calculated based on Equation (4). Unlike the dry core condition, the initial oil content was referenced to the post-water-huff-and-puff state, which served as the baseline for evaluating CO_2 -based recovery in the presence of retained water.

$$R_{wi} = \frac{S_{wi} - S_1}{S_1} \times 100\% \quad (4)$$

Where R_{wi} is the oil recovery of the i -th CO₂ huff-and-puff cycle under water-bearing conditions, %; S_{wi} is the total T₂ signal after the i -th cycle; S_1 is the total T₂ signal after water huff-and-puff.

The oil recovery at each CO₂ huff-and-puff cycle under different water saturation conditions, as well as the incremental recovery per cycle, is shown in Figure 9. Under dry conditions, CO₂ huff-and-puff exhibited a typical pattern of diminishing marginal returns, with recovery increment per cycle following an exponential decline. This behavior aligns with conventional reservoir dynamics, where reservoir energy depletes and the volume of movable oil gradually decreases over successive cycles. In contrast, under water-bearing conditions, the first-cycle recovery was notably lower. This is primarily because a large volume of residual fracturing fluid remained in the fractures during the initial huff-and-puff cycle, resulting in a produced fluid composition dominated by water and limited oil displacement (Liu et al. 2023). However, the incremental improvement in oil recovery across subsequent cycles was significantly greater and showed a slower rate of decline, partially sustaining the effectiveness of CO₂ injection in the later stages. This sustained improvement can be attributed to several mechanisms: the presence of water provides an additional medium for CO₂ dissolution, facilitating stronger micro-scale oil displacement through alternating gas–water interactions. Additionally, the high specific heat capacity of water allows for more uniform and prolonged thermal distribution within the reservoir. Although the absolute oil recovery under water-bearing conditions may be reduced due to dilution by water production, the incremental effect of CO₂ huff-and-puff is notably amplified by the synergistic interaction among gas, liquid, and thermal phases. Ultimately, the presence of water increased the total oil recovery by 9.53% compared to the dry condition.

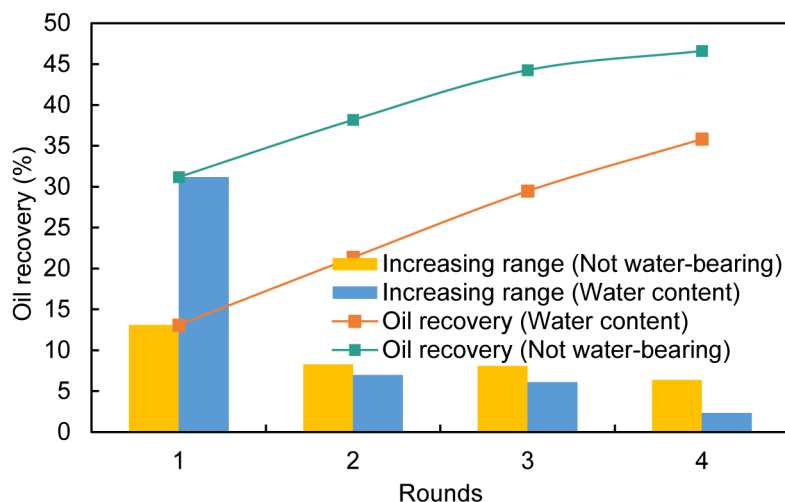


Figure 9—Oil recovery and incremental improvement per CO₂ huff-and-puff cycle under dry and water-bearing conditions.

According to the previously defined classification of shale pore types, the oil recovery from each pore category after four CO₂ huff-and-puff cycles under different water saturation conditions is illustrated in Figure 10. Under dry conditions, oil recovery showed an exponential increase with increasing pore size. This trend is primarily controlled by the relationship between capillary resistance and displacement pressure. In micropores, capillary pressure greatly exceeds the displacement force provided by CO₂ injection, causing oil to remain immobilized in the form of bound water films. In contrast, in macro-pores, the low capillary resistance allows CO₂ to readily displace oil completely. Under water-bearing conditions, oil recovery from micropores was notably lower due to the presence of water hindering the interaction between CO₂ and oil. Additionally, water's surface tension further amplified capillary trapping effects. However, in comparison to dry conditions, oil recovery from small and medium pores improved significantly in the presence of water.

This enhancement is mainly attributed to the water phase acting as a wetting agent, improving pore surface wettability and reducing the adhesive forces at the oil–rock interface. Furthermore, interfacial interactions between water and CO₂ contributed to enhanced micro-scale displacement efficiency.

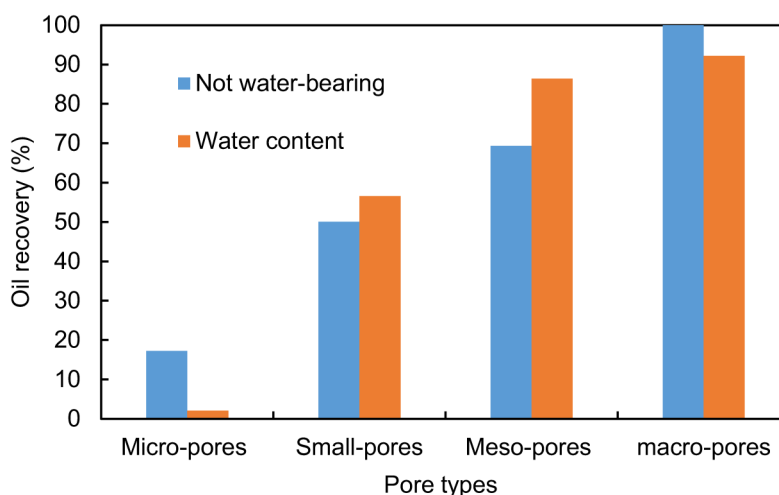


Figure 10—Oil recovery from different pore types under dry and water-bearing conditions after four CO₂ huff-and-puff cycles.

Conclusion

This study investigated the mass transfer and diffusion characteristics of CO₂ as well as its oil recovery and sequestration performance under different temperature and water saturation conditions, using CO₂ convection-diffusion experiments and NMR-assisted CO₂ huff-and-puff tests. The main conclusions are as follows:

1. The CO₂ mass transfer process in porous media can be divided into four stages: interfacial dissolution initiation, convection–diffusion synergy, concentration gradient attenuation, and quasi-steady equilibrium. Elevated temperatures shorten the duration of the initiation phase, accelerate diffusion, and allow the system to reach steady state more rapidly.
2. The combined effects of retained fracturing fluid and formation water form an aqueous-phase barrier that systematically weakens CO₂ mass transfer capacity. As water saturation increases, the pressure decay rate significantly decreases. The aqueous phase increases mass transfer resistance through two mechanisms: blocking pore throats to increase diffusion tortuosity and shifting mass transfer toward gas–liquid interfacial dissolution–desorption.
3. Under low-temperature conditions (<110 °C), oil recovery is limited, whereas high temperatures (>110 °C) significantly enhance recovery. At 140 °C, oil recovery increases linearly with injection cycles, indicating that CO₂ exhibits both efficient displacement and favorable sequestration characteristics at this condition.
4. Oil in macro-pores can be fully recovered at all temperatures. Recovery from small pores improves significantly with increasing temperature, while effective recovery from micropores requires temperatures above 80 °C. A temporary decline in recovery is observed in medium pores around 80 °C due to oil redistribution.
5. Under water-bearing conditions, initial oil recovery is lower; however, the incremental improvements in later cycles are more sustained. The presence of the aqueous phase amplifies the incremental effect of CO₂ huff-and-puff, ultimately increasing overall recovery by 9.53% compared to dry conditions—highlighting the synergistic advantage of three-phase interactions.

Nomenclature

EOR Enhanced oil recovery
NMR Nuclear magnetic resonance

Parameters

$P(t)$ Time-dependent pressure function, MPa
 P_0 Initial pressure, MPa
 D_e Effective diffusion coefficient of CO₂, cm²/s
 A Cross-sectional area of the core, cm²
 t Time, s
 V_g Volume of the CO₂ container, mL
 L Length of the core, cm
 m Slope of the linear fit of $\ln P(t)$ versus time t
 R_i Oil recovery of the i -th CO₂ huff-and-puff cycle, %
 S_i Total T₂ signal after the i -th cycle
 S_0 Total T₂ signal in the original state
 R_{wi} Oil recovery of the i -th CO₂ huff-and-puff cycle under water-bearing conditions, %
 S_{wi} Total T₂ signal after the i -th cycle
 S_1 Total T₂ signal after water huff-and-puff

References

- Alafnan, Saad, Aljawad, Murtada, Glatz, Guenther et al 2020. Sustainable Production from Shale Gas Resources through Heat-Assisted Depletion. *Sustainability* **12** (5): 2145.
- Chen, Zheng, Su, Yuliang, Li, Lei et al 2025. Experimental exploration of CO₂ dissolution, diffusion, and flow characteristics in bound water conditions during oil displacement and sequestration. *Fuel* **386**: 134249.
- Goodman, Angela, Kutchko, Barbara, Sanguinito, Sean et al 2021. Reactivity of CO₂ with Utica, Marcellus, Barnett, and Eagle Ford Shales and impact on permeability. *Energy & Fuels* **35** (19): 15894–15917.
- Guo, Yujia, Shi, Yue, and Mohanty, Kishore. 2025. Comparison of the Effectiveness of CO₂ and Hydrocarbon Gas for Shale Huff'n'Puff. *SPE Journal*: 1–16.
- Li, Bo, Li, Gang, Li, Xiao-Sen et al 2014. The use of heat-assisted antigravity drainage method in the two horizontal wells in gas production from the Qilian Mountain permafrost hydrate deposits. *Journal of Petroleum Science and Engineering* **120**: 141–153.
- Li, Lei, Wang, Chengwei, Li, Dongsheng et al 2019. Experimental investigation of shale oil recovery from Qianjiang core samples by the CO₂ huff-n-puff EOR method. *RSC advances* **9** (49): 28857–28869.
- Li, Lirong, Lim, Hwan Suk, and Kang, Yong Tae. 2019. Mass transfer characteristics for CO₂ absorption in nanoabsorbents. *International Journal of Heat and Mass Transfer* **129**: 650–659.
- Liu, Junrong, Li, Hangyu, Liu, Shuyang et al 2023. Investigating the Impact of Aqueous Phase on CO₂ Huff 'n' Puff in Tight Oil Reservoirs Using Nuclear Magnetic Resonance Technology: Stimulation Measures and Mechanisms. *SPE Journal* **28** (06): 3324–3340.
- Lu, Yiyu, Tian, Rongrong, Tang, Jiren et al 2021. Investigating the mineralogical and chemical effects of CO₂ injection on shale wettability at different reservoir temperatures and pressures. *Energy & Fuels* **35** (18): 14838–14851.
- Mo, Fei, Qi, Zhilin, Huang, Xiaoliang et al 2022. Knudsen diffusion in pore-scale water-bearing shales: modelling and case study. *Journal of Petroleum Science and Engineering* **214**: 110457.
- Nikkhou, Fatemeh, Keshavarz, Peyman, Ayatollahi, Shahab et al 2015. Evaluation of interfacial mass transfer coefficient as a function of temperature and pressure in carbon dioxide/normal alkane systems. *Heat and Mass Transfer* **51**: 477–485.
- Ning, Yanrui and Tura, Ali. 2023. Economic and operational investigation of CO₂ sequestration through enhanced oil recovery in unconventional reservoirs in Colorado, USA. *Geoenergy Science and Engineering* **226**: 211820.
- Nsima, Kaloucha Kanga, Yu, Long, Wang, Fei et al 2024. Exhaustive Review of CO₂ Sequestration in Depleted Hydrocarbon Reservoirs: Recent Advances, Challenges and Future Prospects. *Energy & Fuels* **38** (22): 21701–21734.
- Nunn, Jacob A, Xiang, Yan, and Al, Tom A. 2018. Investigation of partial water saturation effects on diffusion in shale. *Applied Geochemistry* **97**: 93–101.

- Pu, Wanfen, Wei, Bing, Jin, Fayang et al 2016. Experimental investigation of CO₂ huff-n-puff process for enhancing oil recovery in tight reservoirs. *Chemical Engineering Research and Design* **111**: 269–276.
- Shi, Yang, Weng, Dingwei, Cai, Bo et al 2024. Flow and Heat Transfer of Shale Oil Reservoir during CO₂ Enhanced Pyrolysis: A Pore-Scale Modeling. *Processes* **12** (8): 1694.
- Song, Chengyao and Yang, Daoyong. 2017. Experimental and numerical evaluation of CO₂ huff-n-puff processes in Bakken formation. *Fuel* **190**: 145–162.
- Sui, Hongguang, Zhang, Fengyun, Wang, Ziqiang et al 2020. Effect of kerogen maturity, water content for carbon dioxide, methane, and their mixture adsorption and diffusion in kerogen: A computational investigation. *Langmuir* **36** (33): 9756–9769.
- Tian, Kangjian, Wang, Tianyu, Ma, Zhengchao et al 2024. Dynamic Behaviors of CO₂ Enhanced Shale Oil Flow in Nanopores by Molecular Simulation. *SPE Journal*: 1–17.
- Vengosh, Avner, Jackson, Robert B, Warner, Nathaniel et al 2014. A critical review of the risks to water resources from unconventional shale gas development and hydraulic fracturing in the United States. *Environmental science & technology* **48** (15): 8334–8348.
- Wan, Tao, Chen, Sen, Wu, Xiaolei et al 2024. An investigation on the pyrolysis of low-maturity organic shale by different high temperature fluids heating. *Fuel* **374**: 132498.
- Wang, Qiang, Chen, Xi, Jha, Awadhesh N et al 2014. Natural gas from shale formation—the evolution, evidences and challenges of shale gas revolution in United States. *Renewable and Sustainable Energy Reviews* **30**: 1–28.
- Xia, Zhizeng, Yin, Hongjun, Wang, Xuewu et al 2024. Surfactant slug assisting CO₂ huff and puff in enhancing shale oil reservoir recovery. *Physics of Fluids* **36** (1).
- Xie, Qichao, Song, Peng, Cao, Likun et al 2023. Investigation of Key Controlling Factors and Applicability Boundaries of Natural Gas Injection for Shale Oil Development: A Case Study of Chang 7 Reservoir, Ordos Basin, China. *Energies* **16** (21): 7377.
- Yan, SONG, Zhuo, LI, Qun, LUO et al 2017. Progress and development trend of unconventional oil and gas geological research. *Petroleum Exploration and Development* **44** (4): 675–685.
- Yang, Chaodong and Gu, Yongan. 2006. Accelerated mass transfer of CO₂ in reservoir brine due to density-driven natural convection at high pressures and elevated temperatures. *Industrial & engineering chemistry research* **45** (8): 2430–2436.
- Zhao, Jia, Yao, Chuanjin, Cheng, Tianxiang et al 2024. Heating-Assisted CO₂ Huff-n-Puff for Improved CO₂ Utilization: Further Discussion of the EOR Mechanism and Enhancing Carbon Sequestration in Shale Reservoirs. *Energy & Fuels* **38** (5): 3937–3951.
- Zhao, Yongliang, Feng, Yanhui, and Zhang, Xinxin. 2016. Molecular simulation of CO₂/CH₄ self-and transport diffusion coefficients in coal. *Fuel* **165**: 19–27.
- Zheng, Sixu, Sun, Huijuan, and Yang, Daoyong. 2016. Coupling heat and mass transfer for determining individual diffusion coefficient of a hot C₃H₈–CO₂ mixture in heavy oil under reservoir conditions. *International Journal of Heat and Mass Transfer* **102**: 251–263.
- Zuloaga, Pavel, Yu, Wei, Miao, Jijun et al 2017. Performance evaluation of CO₂ Huff-n-Puff and continuous CO₂ injection in tight oil reservoirs. *Energy* **134**: 181–192.